

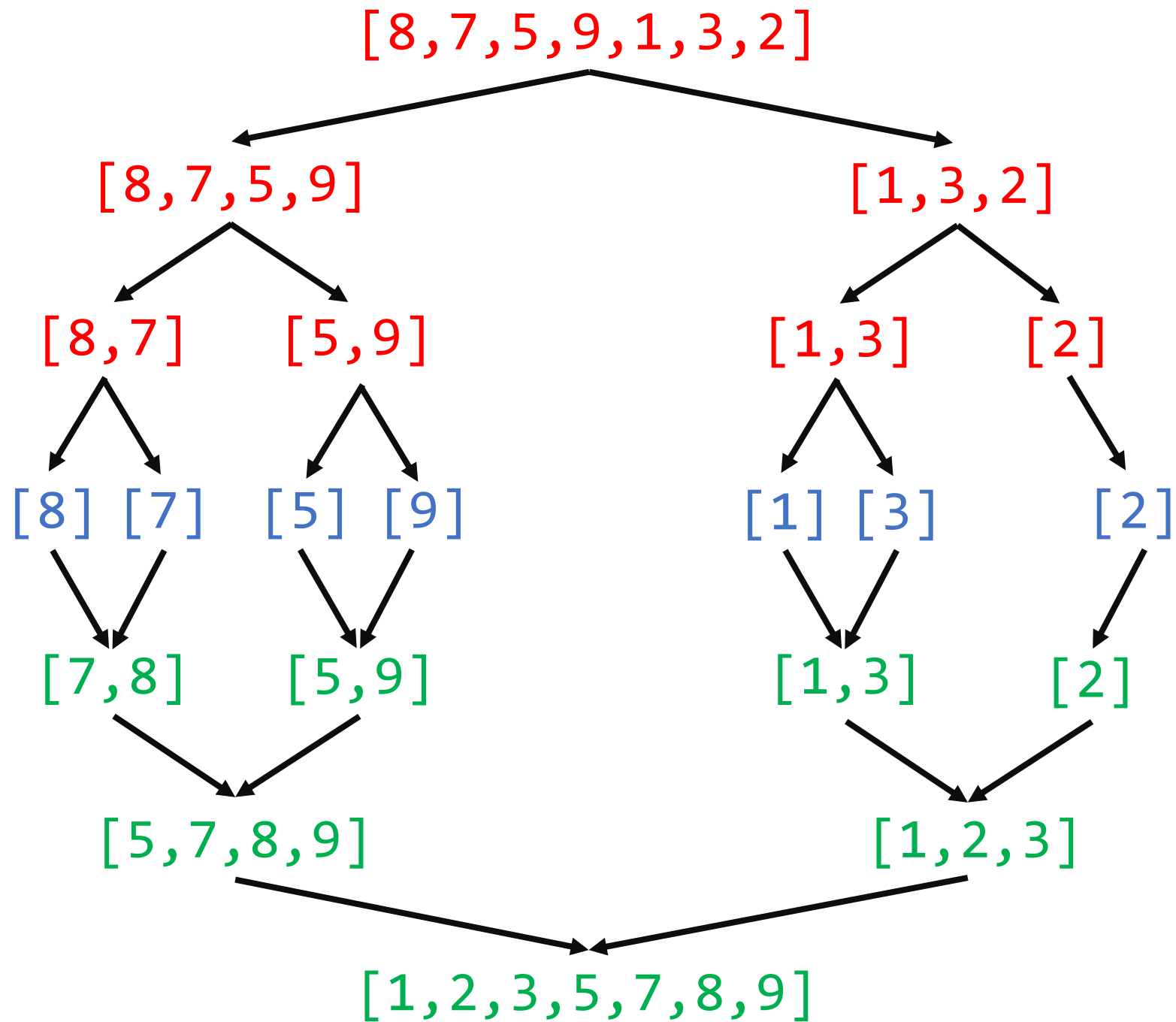
Merge Sort

Animation

Main Idea

- Divide and Conquer
- Divide the unsorted list into n sublists, each containing one item (a list of one element is item sorted).
- Repeatedly merge sublists to produce new sorted sublists until there is only one sublist remaining. This will be the sorted list.

Visualising the whole process



Visualising the recursive calls

[8,7,5,9,1,3,2]

Start sort of list

[8, 7, 5, 9, 1, 3, 2]



[8, 7, 5, 9]

Sort left half of list

[8,7,5,9,1,3,2]

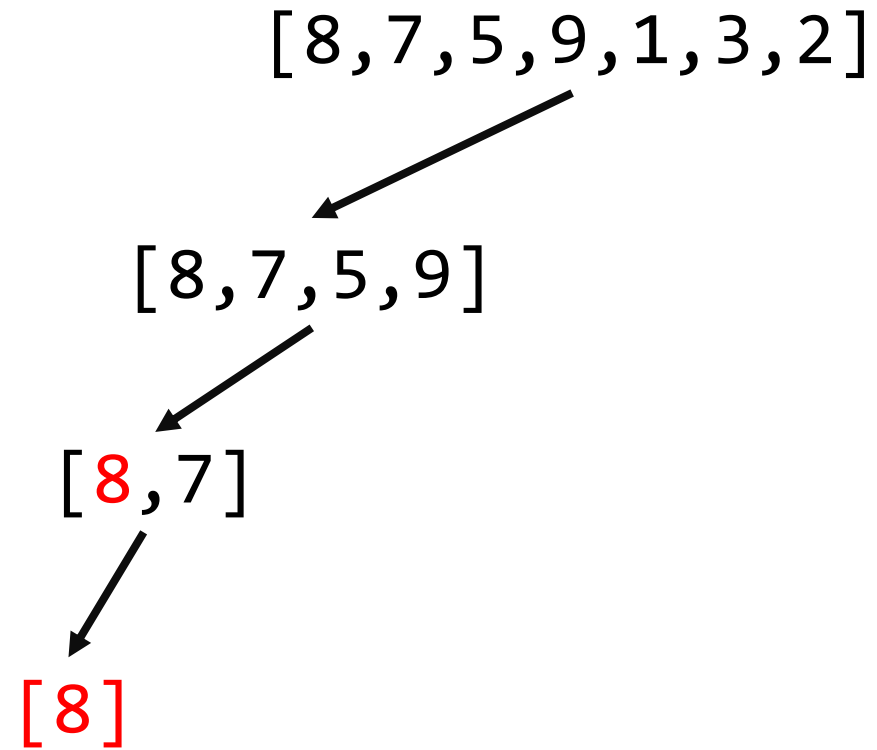


[8,7,5,9]

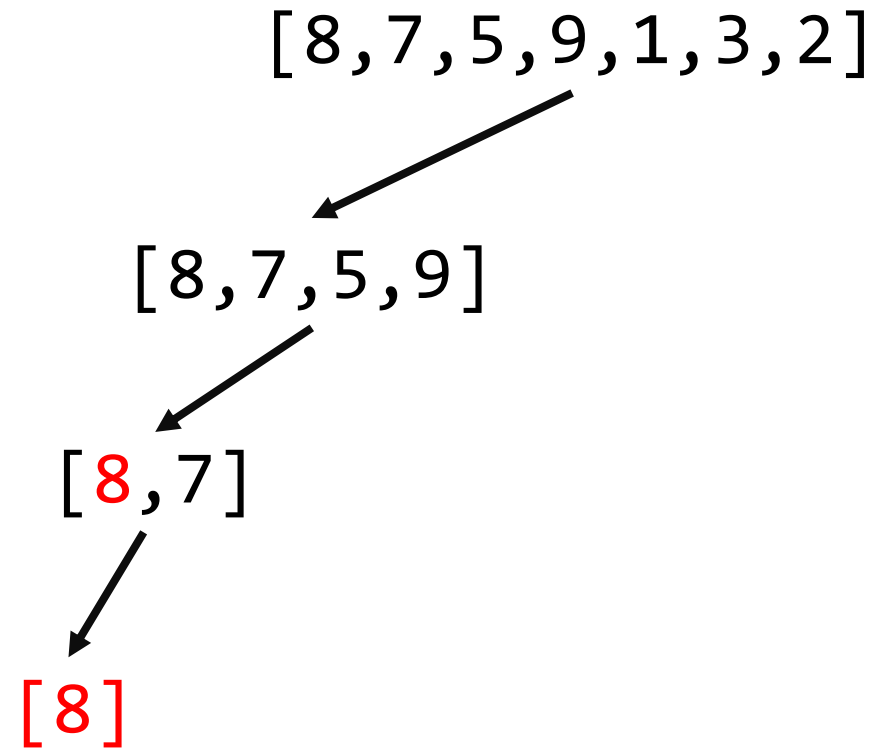


[8,7]

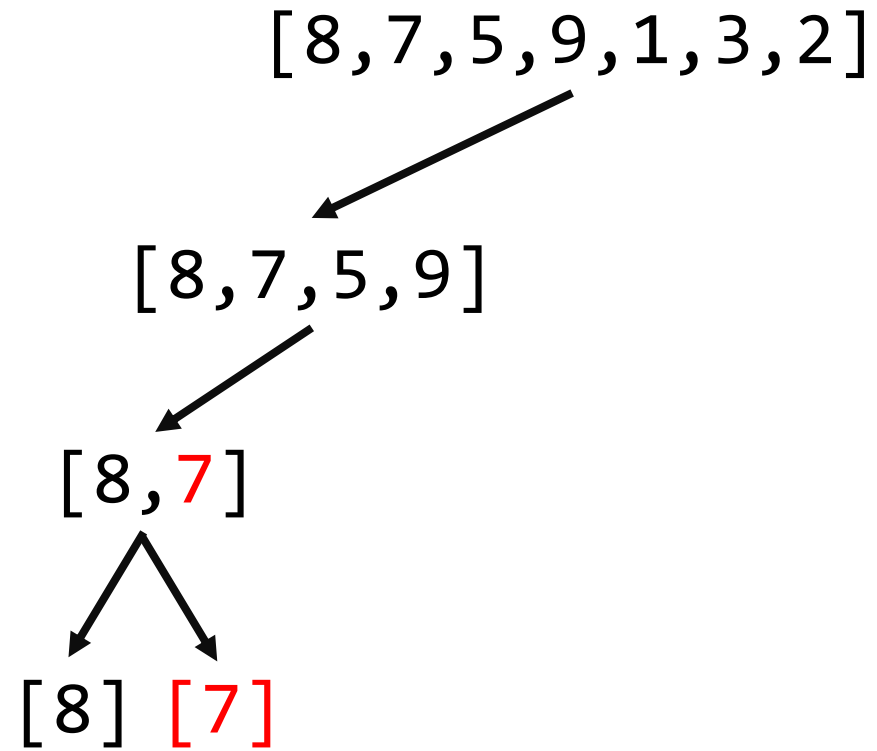
Sort left half of list



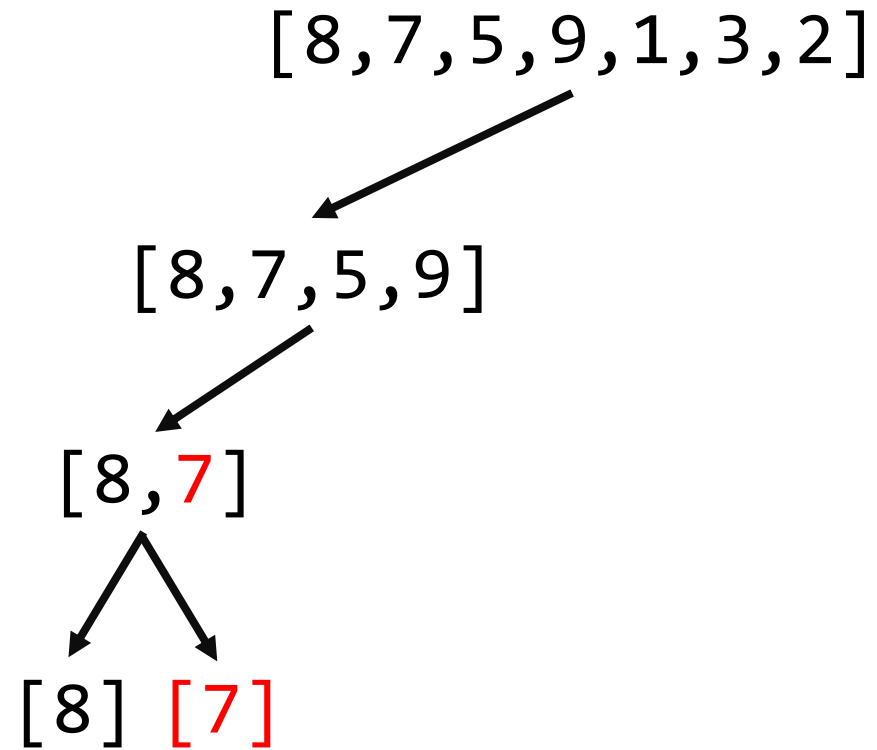
Sort left half of list



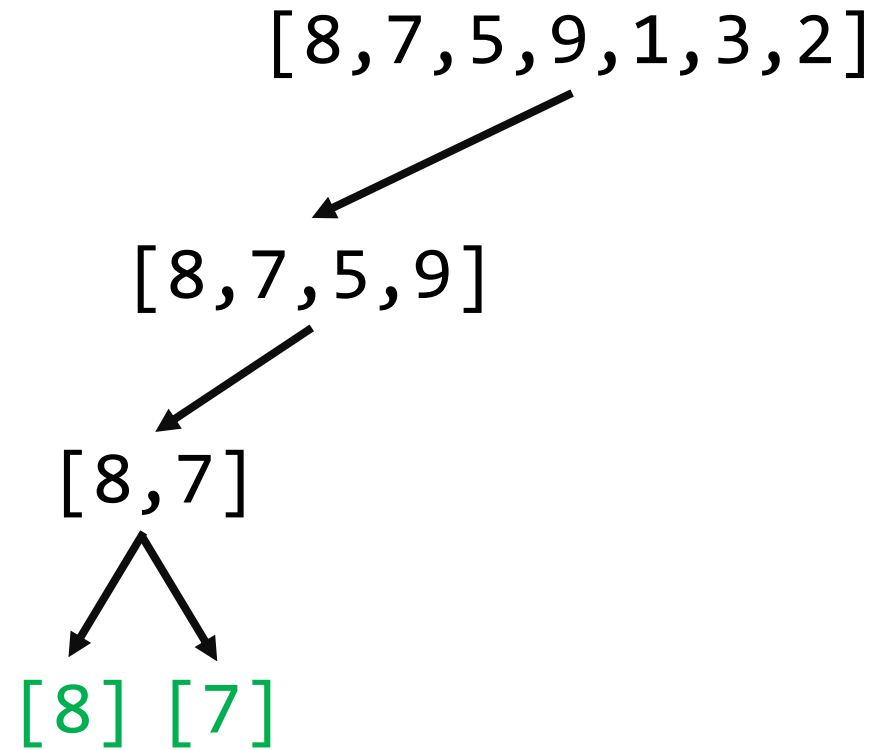
Base Case! Length of list is 1



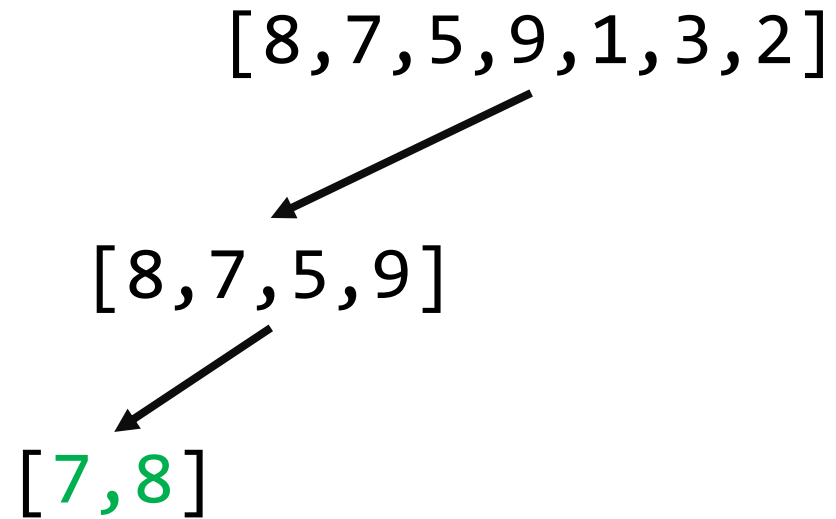
Sort right half of list



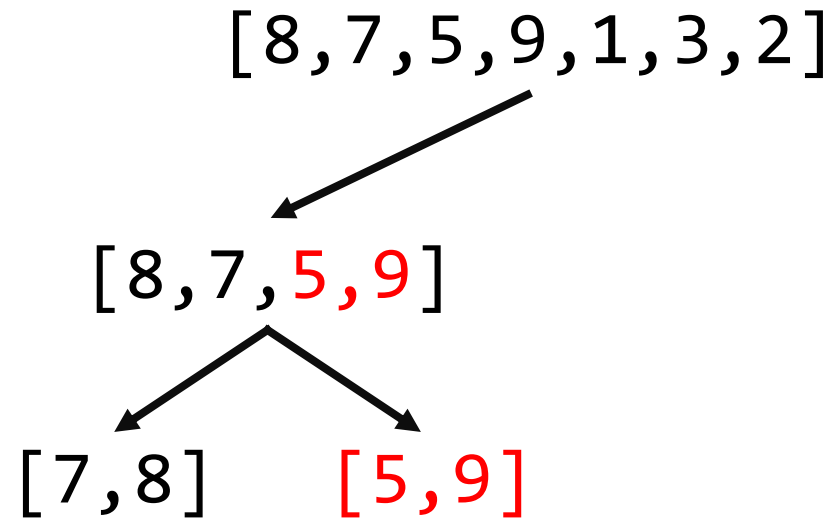
Base Case! Length of list is 1



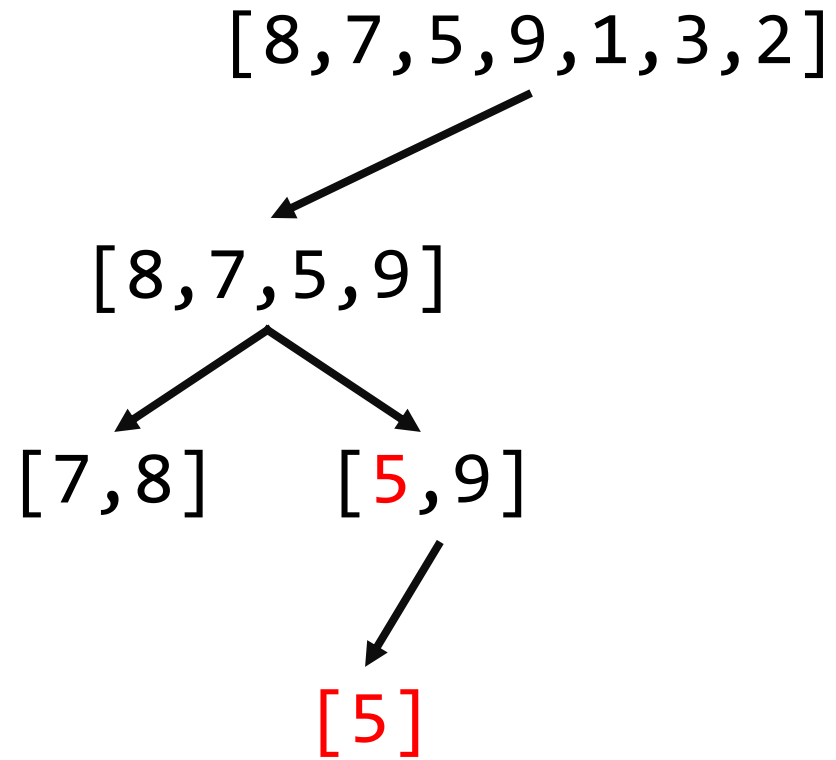
We now merge the two lists



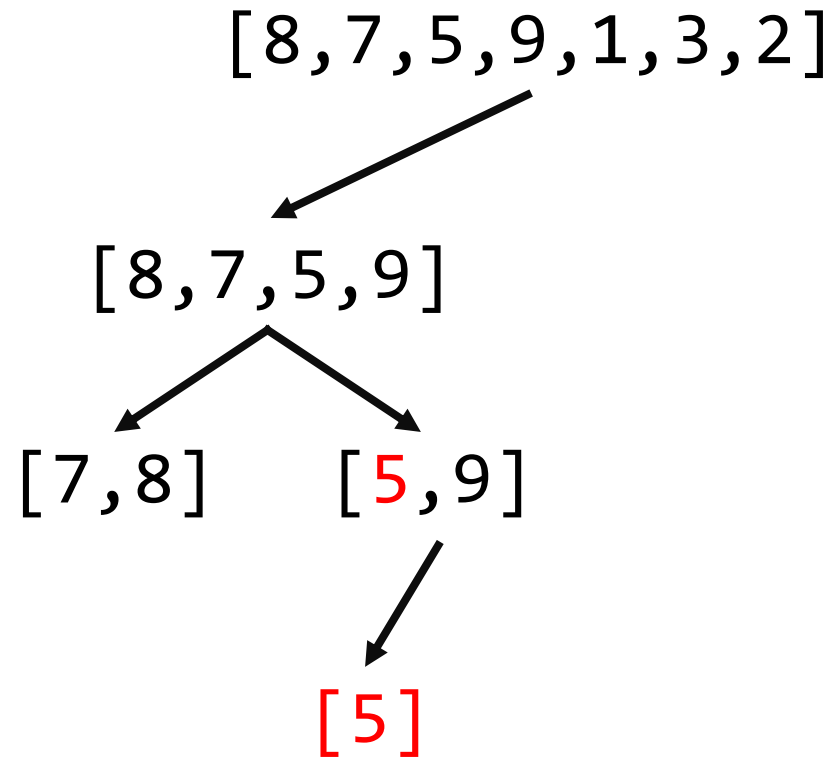
We now merge the two lists



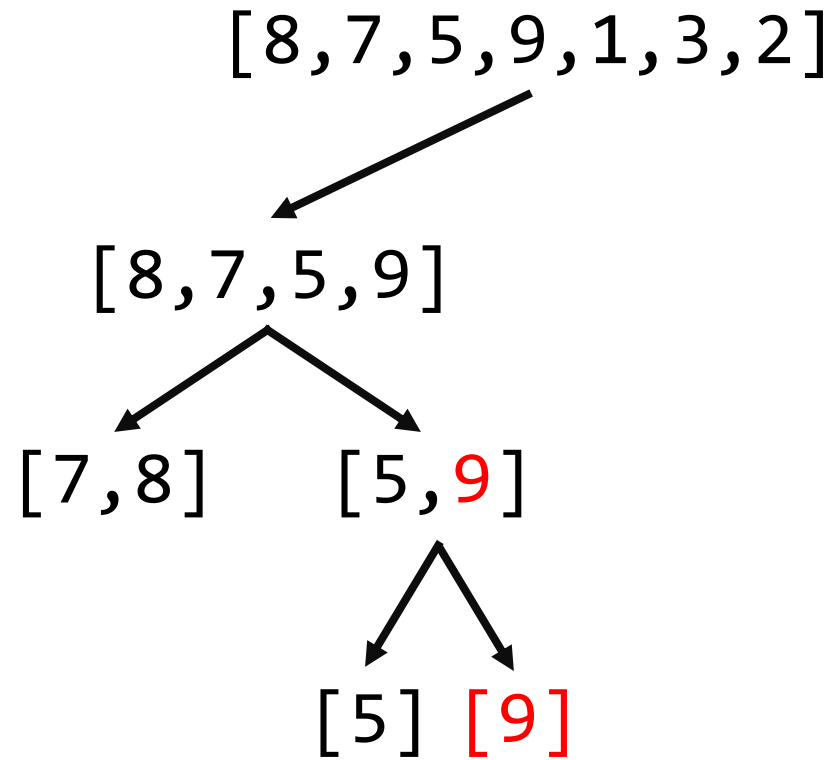
Sort right half of list



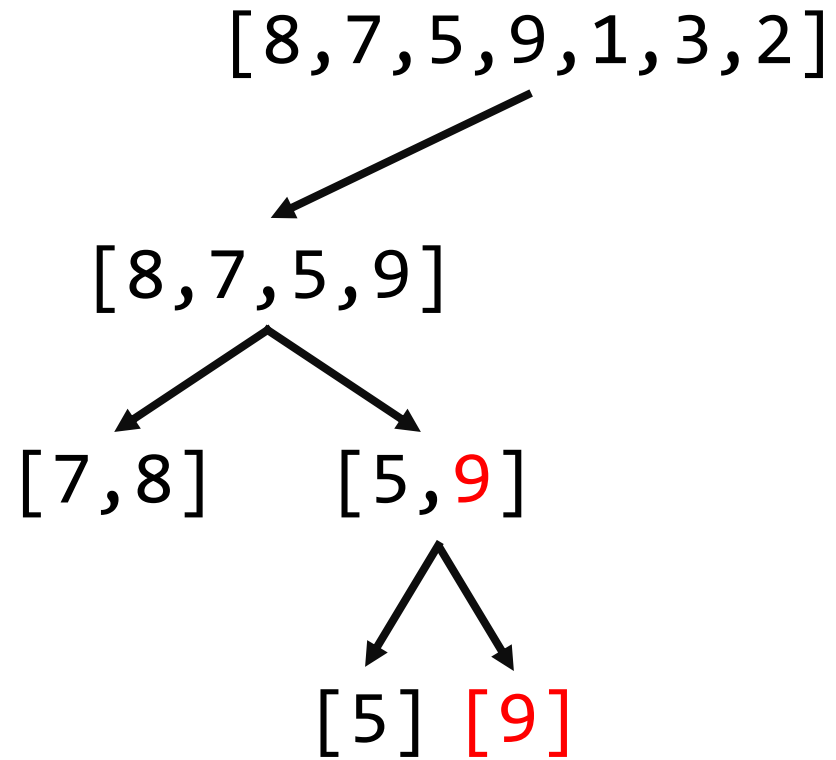
Sort left half of list



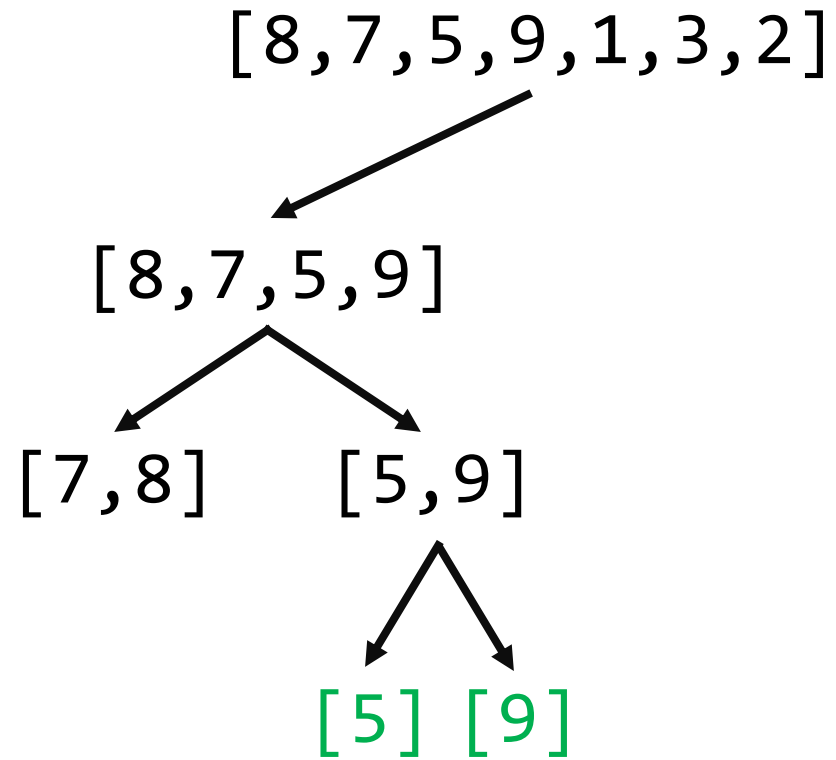
Base Case! Length of list is 1



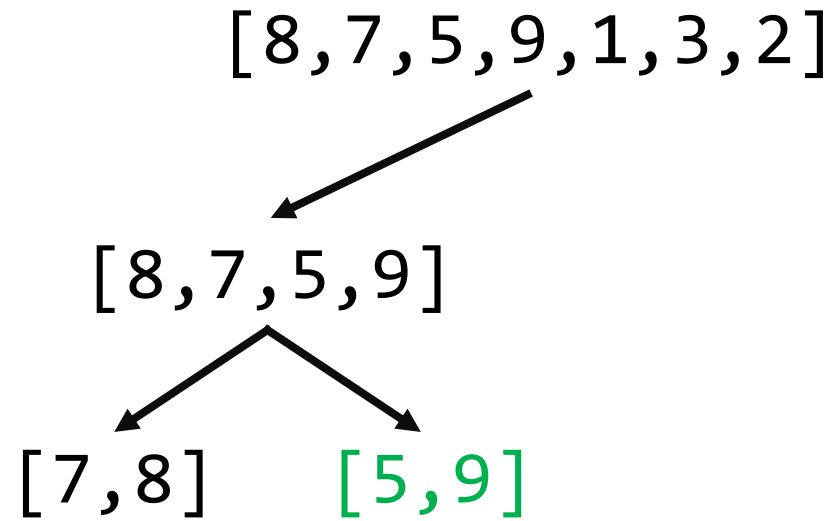
Sort right half of list



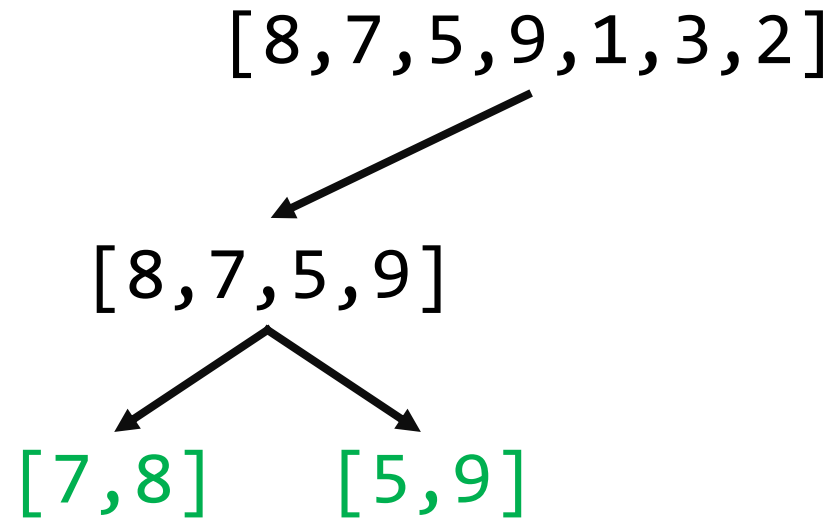
Base Case! Length of list is 1



We now merge the two lists



We now merge the two lists

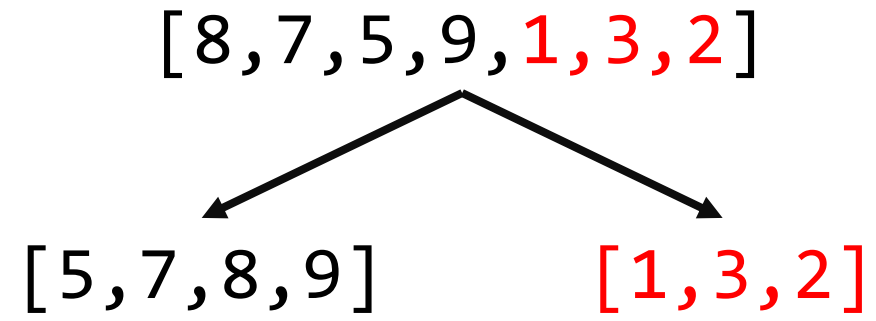


We now merge the two lists

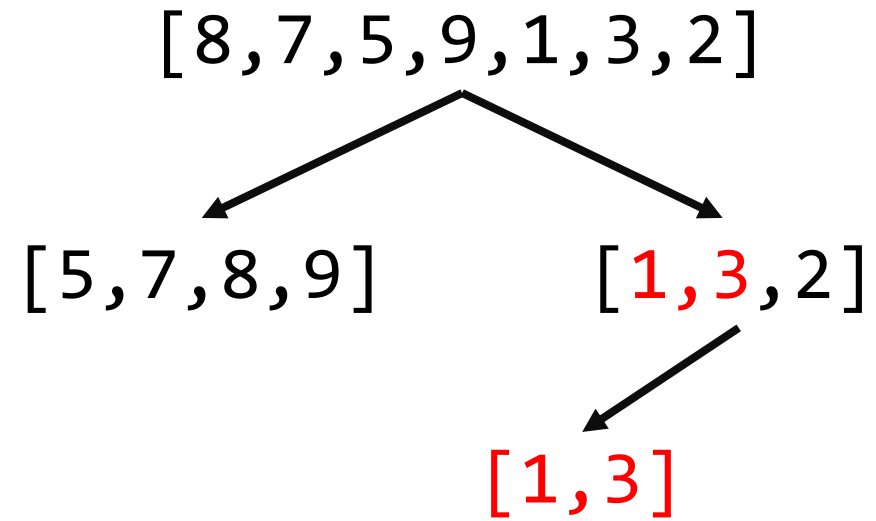
[8,7,5,9,1,3,2]



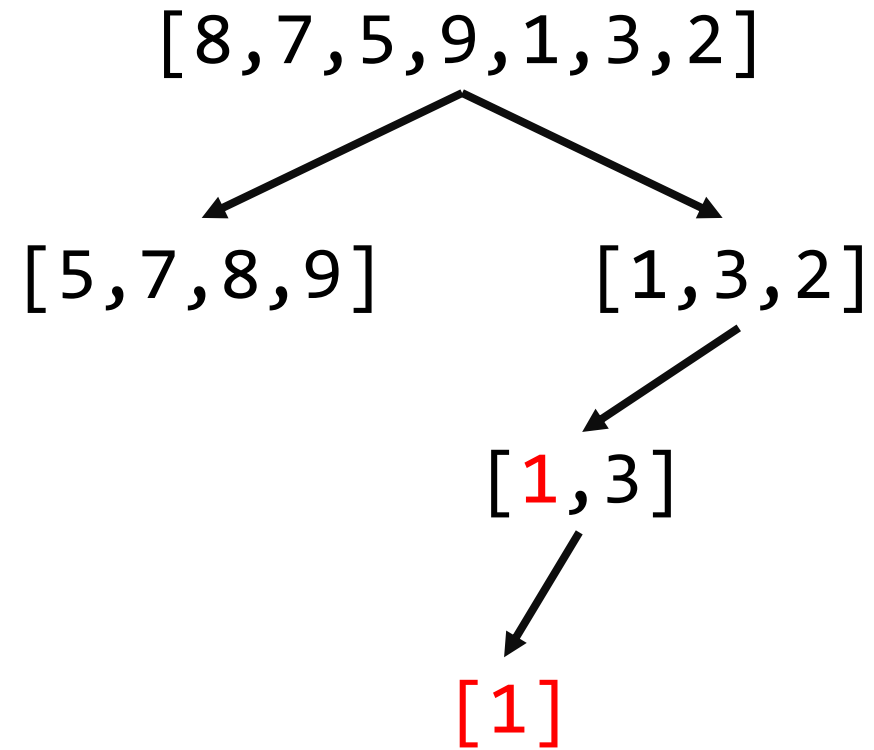
[5,7,8,9]



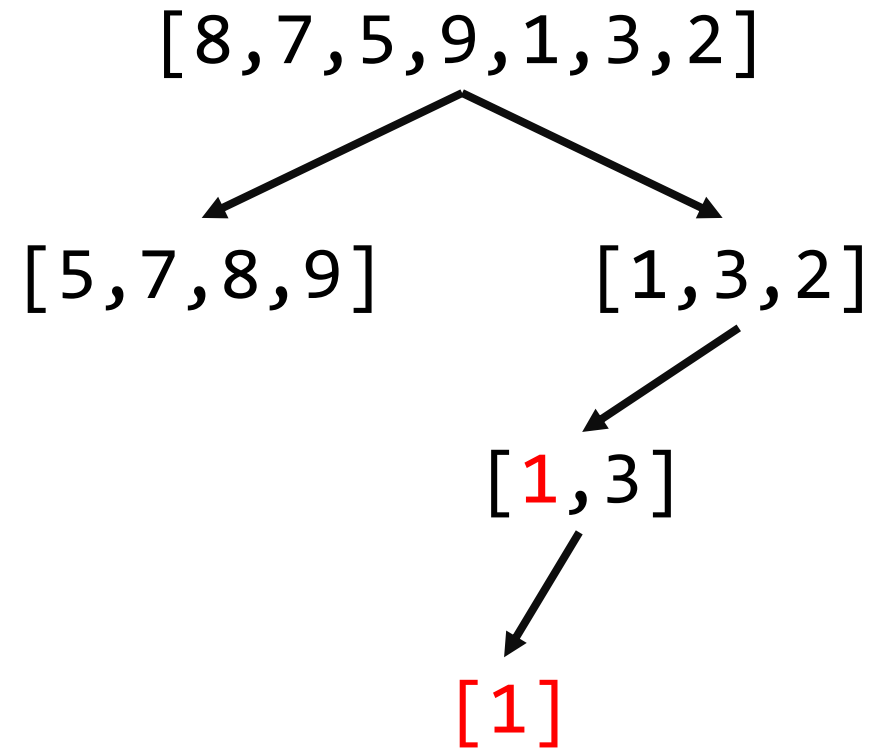
Sort right half of list



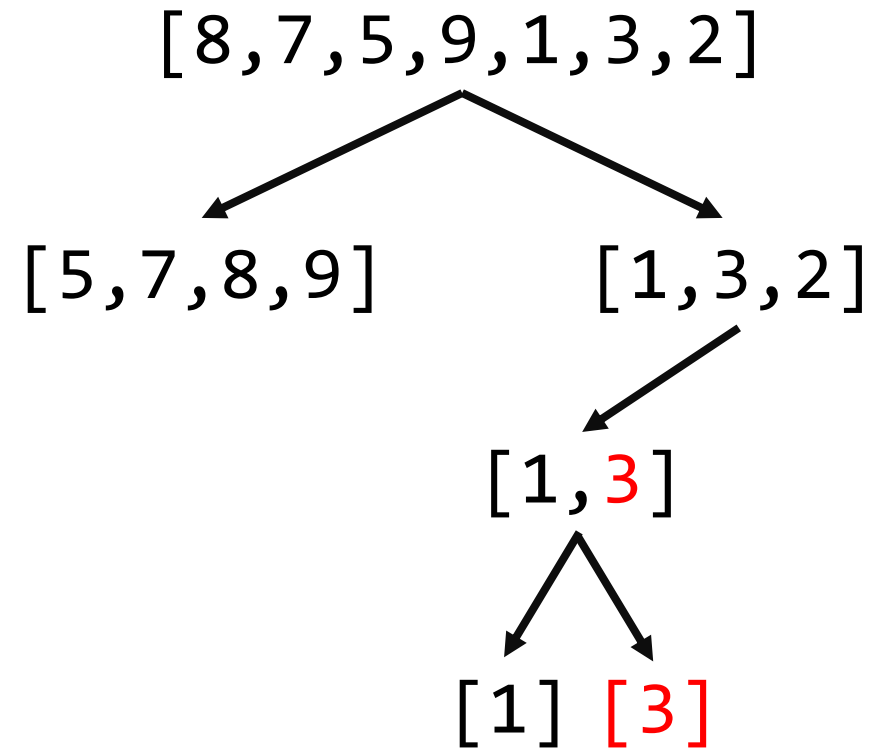
Sort left half of list



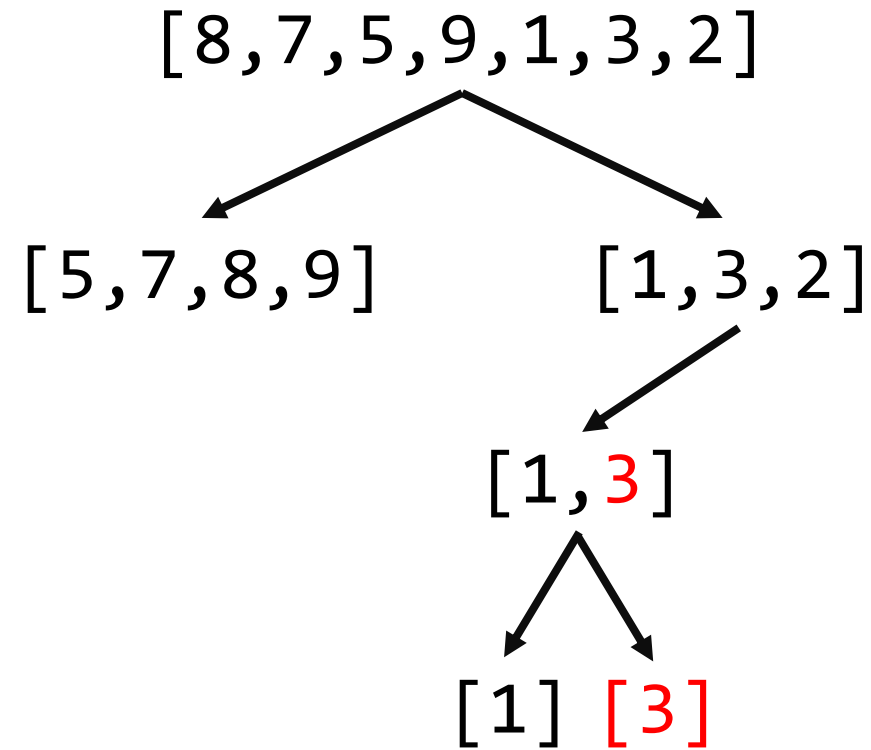
Sort left half of list



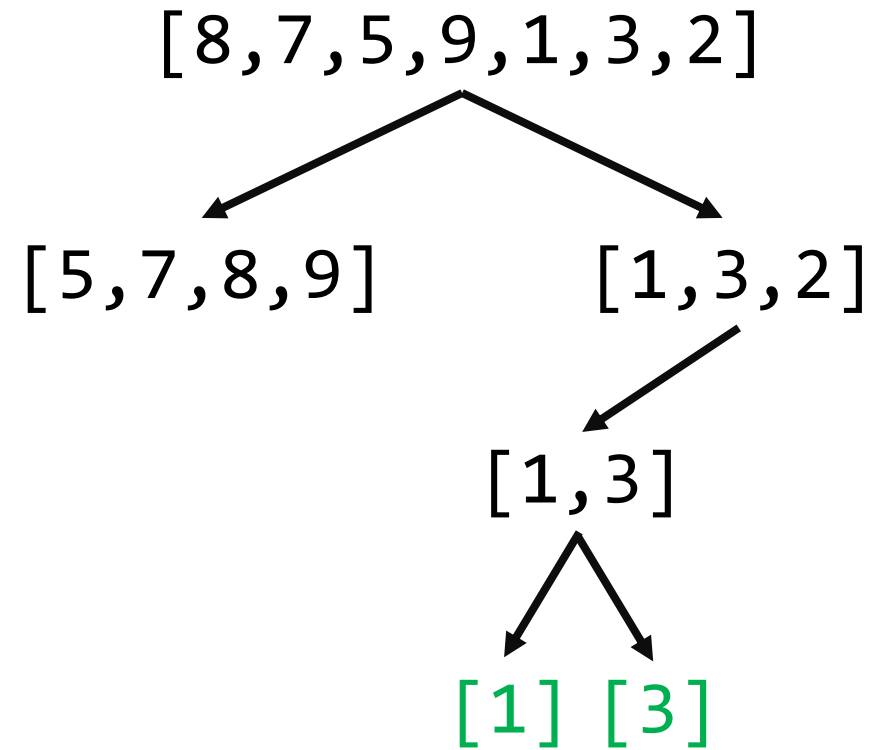
Base Case! Length of list is 1



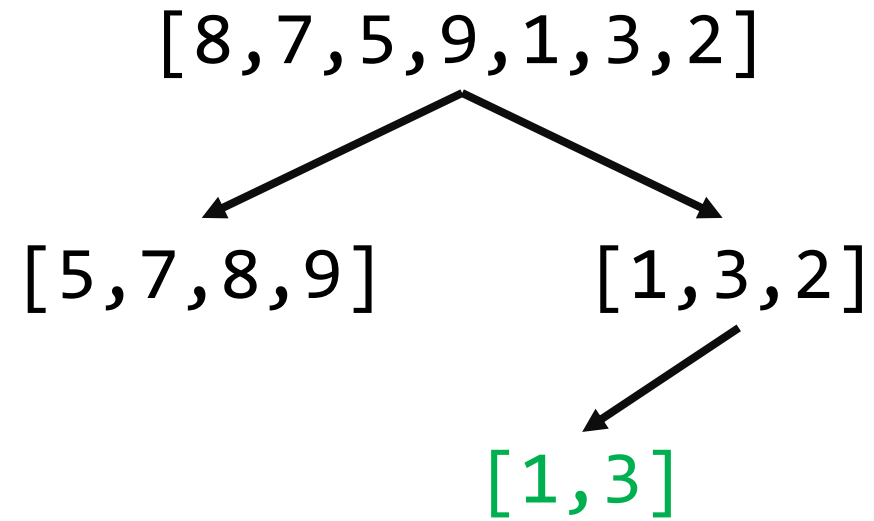
Sort right half of list



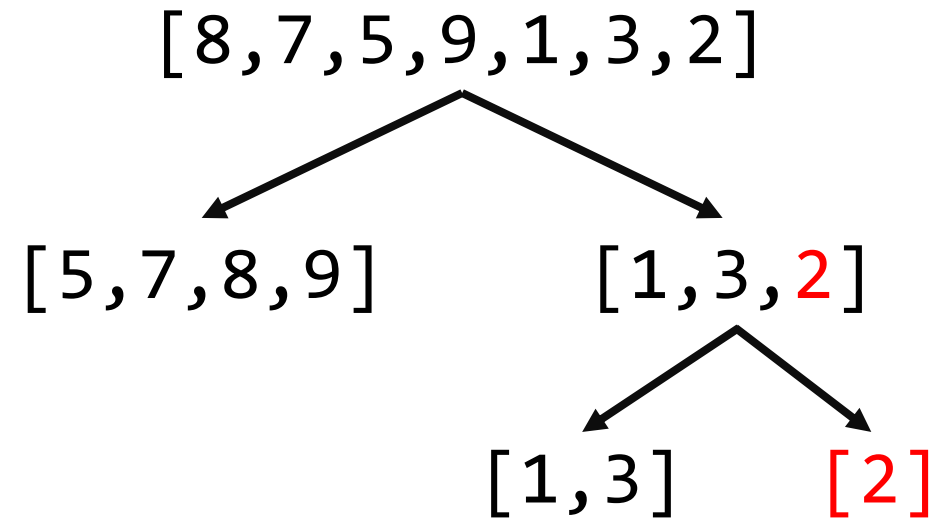
Base Case! Length of list is 1



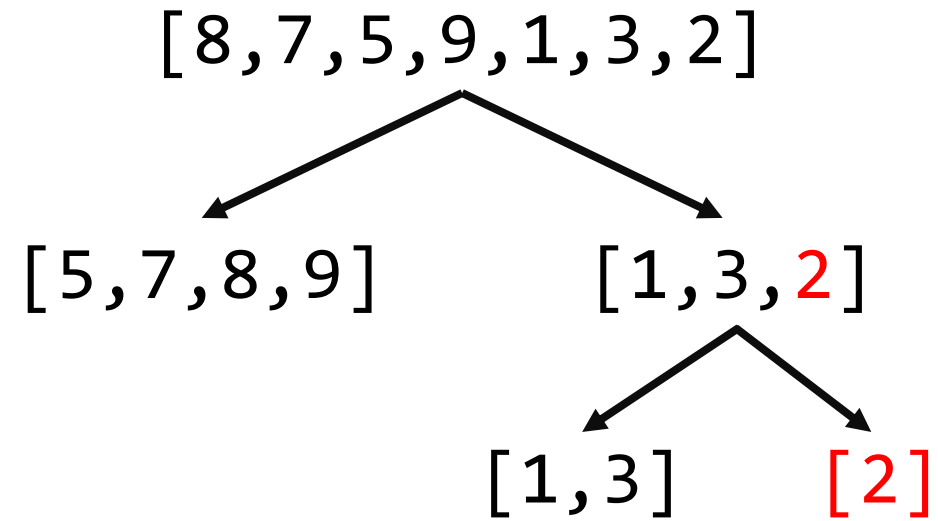
We now merge the two lists



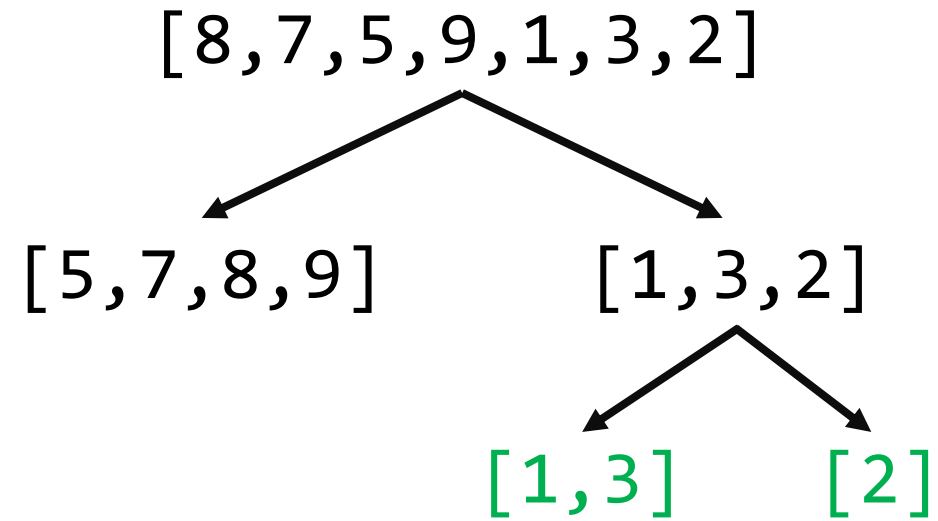
We now merge the two lists



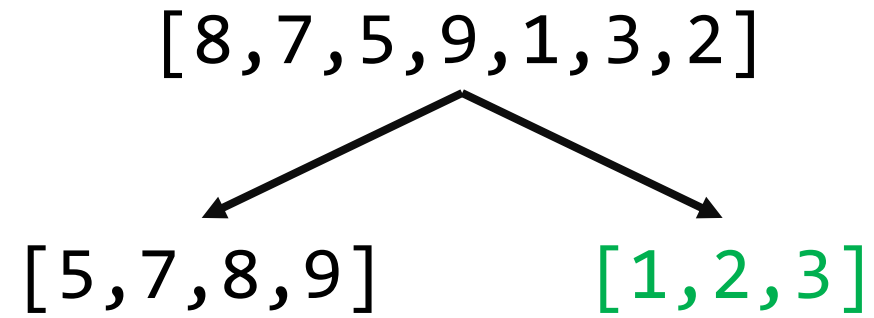
Sort right half of list



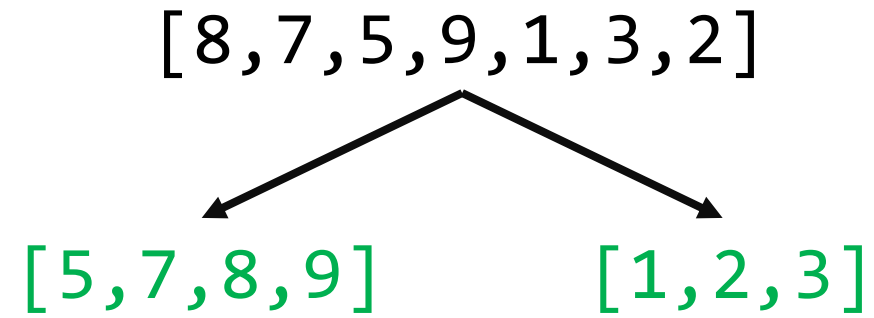
Base Case! Length of list is 1



We now merge the two lists



We now merge the two lists



We now merge the two lists

[1,2,3,5,7,8,9]

We now merge the two lists

```
[1,2,3,5,7,8,9]
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The list is now sorted!